Introducción a la secuenciación genómica de microorganismos de importancia en salud pública: Tecnología Illumina

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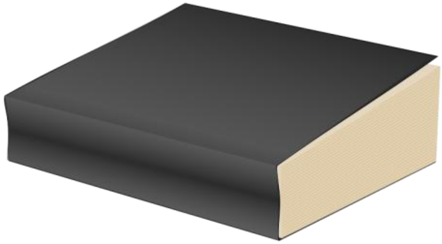
Bases del análisis de WGS

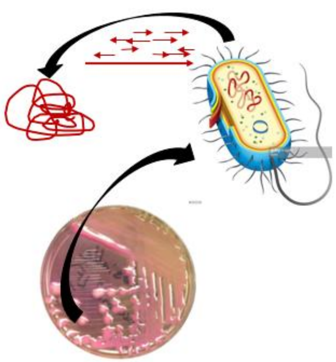


Para NGS se empieza con un genoma (libro completo), se descompone en millones de pedazos (sequence reads) y se trata de volver a construir utilizando algoritmos matemáticos.

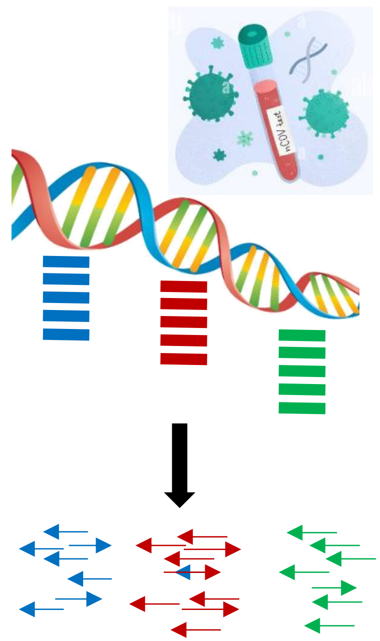
2016 Carleton, H. CDC

NGS Applications for Infectious Diseases

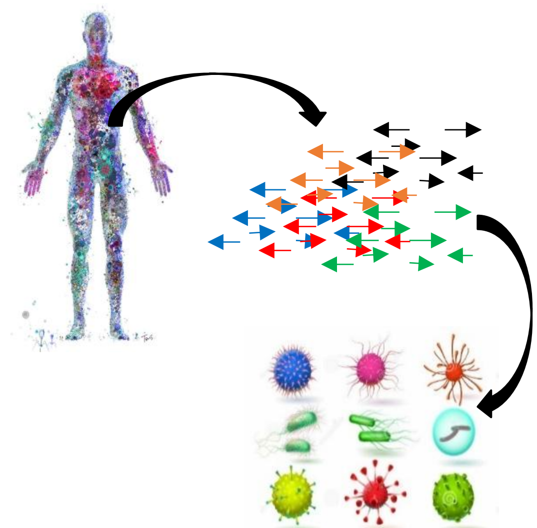
A. Whole Genome Sequencing



B. Targeted NGS (tNGS)

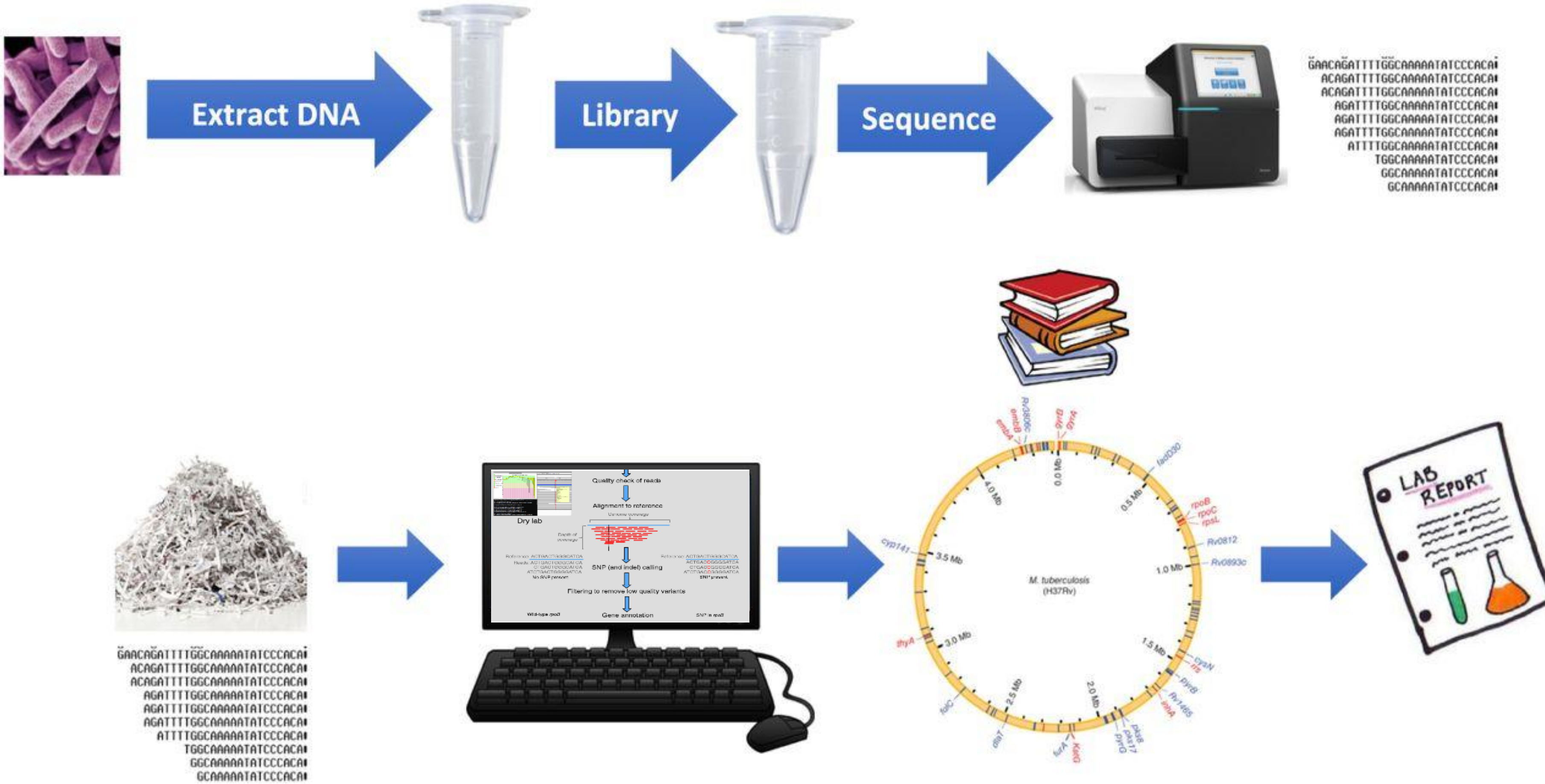


C. Metagenomic NGS (mNGS)



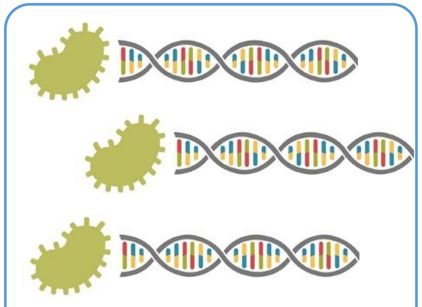
Estrategias para secuenciación de microorganismos para diagnóstico y seguimiento de enfermedades infecciosas

A. Whole Genome Sequencing

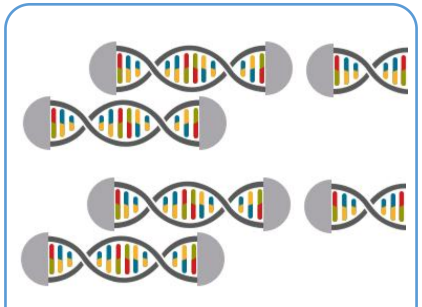


Escuyer, V. Use of Whole-Genome Sequencing for Detection of Antimicrobial Resistance: *Mycobacterium tuberculosis*, a Model Organism. Society for Clinical Laboratory Science Apr 2019, 32 (2) 85-92; **DOI:** 10.29074/ascls.2019001784

El proceso de secuenciación del genoma

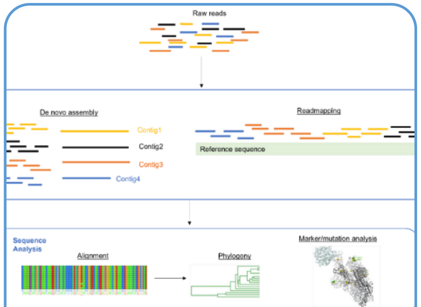


Extracción del genoma 1



Preparación de bibliotecas 2

Secuenciación 3



Análisis bioinformático 4

Adaptado de: [What is Genomic Sequencing? | AMD | CDC]

Extracción del genoma 1

DNA extraction and purification

Key quality indicators and quality control considerations:

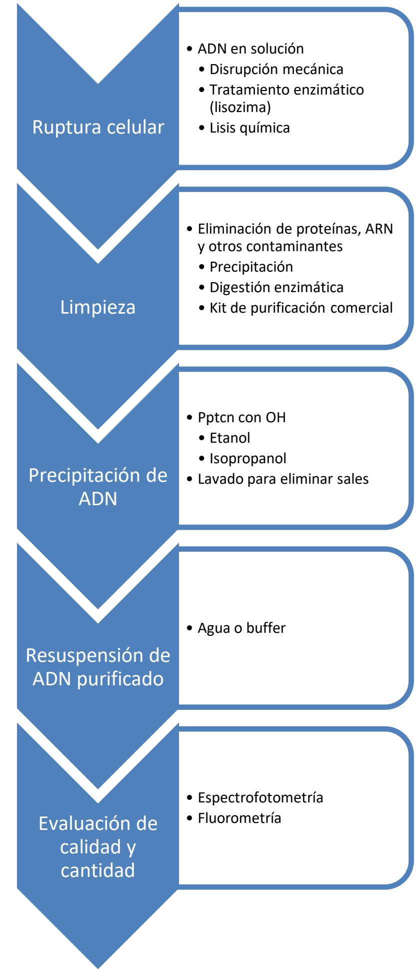
- Specimen/culture sample quality: adequate sample purity (with consideration for other organism DNA, human DNA or inhibitors in the sample)
- Specimen/sample quantity: Sufficient starting material (with consideration for targeted NGS or WGS application)
- DNA quality: Adequate DNA purity (spectrophotometer, $OD_{260/280}$ 1.8-2.0; $OD_{260/230}$ 2.0-2.2)
- DNA quantity: Sufficient starting material (fluorescent detection system, e.g. a Qubit® fluorometer or qPCR); Adequate volume
- DNA size and integrity (agarose gel electrophoresis or microfluidic instruments, including Bioanalyzer)

NOTES:

✓ DNA quantity and quality requirements are dependent on the sequencing application and platform. DNA Quantity/quality assessment must be always conducted after extraction step.

✓ DNA extraction procedures require adequate biosafety containment level when starting from direct specimens or cultured isolates, including appropriate safety equipment and work practices

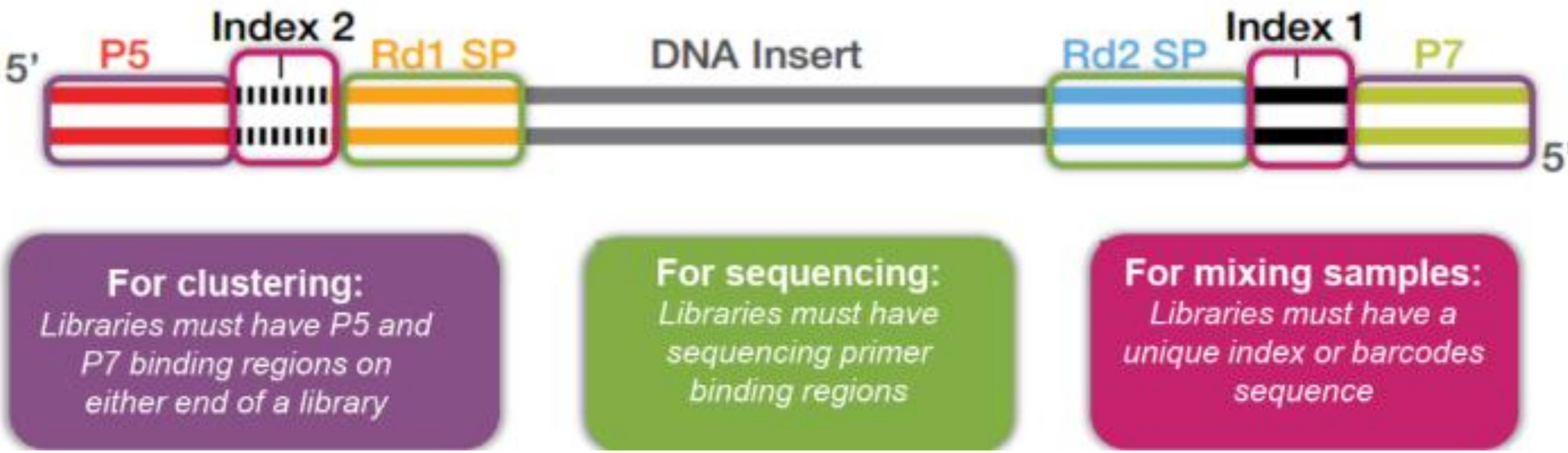
Supporting information: *World Health Organization; 2018 (WHO/CDS/TB/2018.19)*



ADN intacto en alta concentración* y con baja cantidad de contaminantes:

- 260/280: 1,8-2,0 (valores bajos indican contaminación de proteína)
- 260/230: 2,0-2,2 (mide la pureza de los ácidos nucleicos y es un indicador de la contaminación por sales, fenol o carbohidratos)
- Cn para DNAPrep:
    - Min 4 ng/ul
    - Ideal > 10 ng/ul

Illumina Bead-Linked Transposome Chemistry



Preparación de bibliotecas para secuenciación: Illumina DNAprep kit

TAGMENTACIÓN

+20 µL de H_2O
+ 10 µL de ADN*

Pipetear 5-10x

*Cnc debe estar entre 100-500 ng
Mínimo que se agrega de ADN es 2 µL

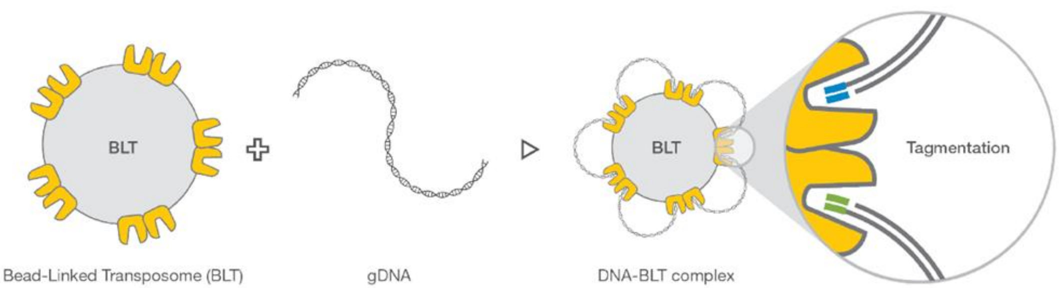
→ Preparar Mix de tagmentación
10 µl TB1
10 µl BLT

*Vortex BLT mínimo 10 seg
Vortex TB1 y spin

→ + 20 µL MasterMix tagmentación

Pipetear 5-10x

→ Termociclador 55°C por 15 min
FLEX1
(volumen es de 50 µl)



Bruinsma, S., Burgess, J., Schlingman, D. *et al.* Bead-linked transposomes enable a normalization-free workflow for NGS library preparation. *BMC Genomics* **19,** 722 (2018). https://doi.org/10.1186/s12864-018-5096-9

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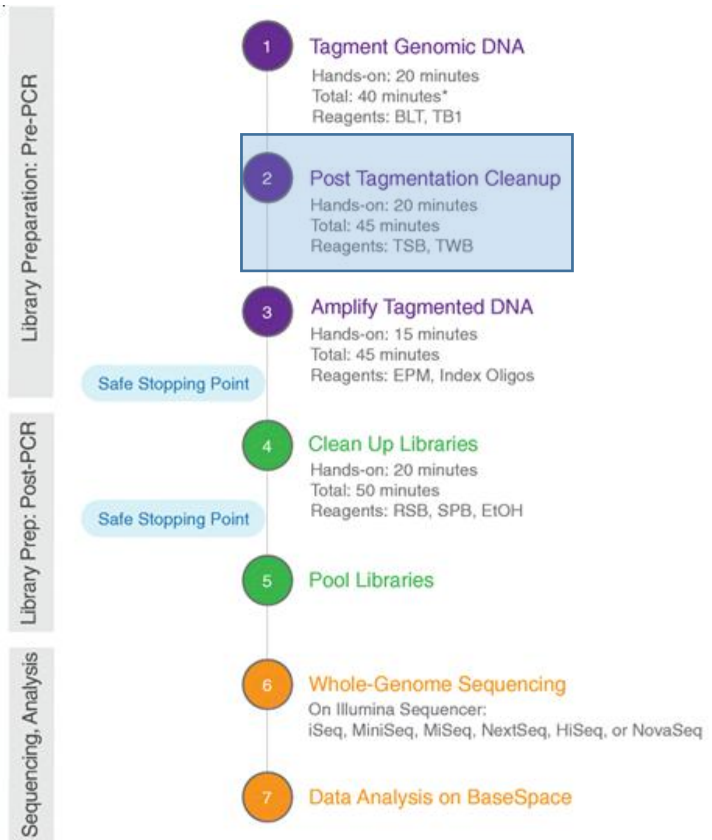

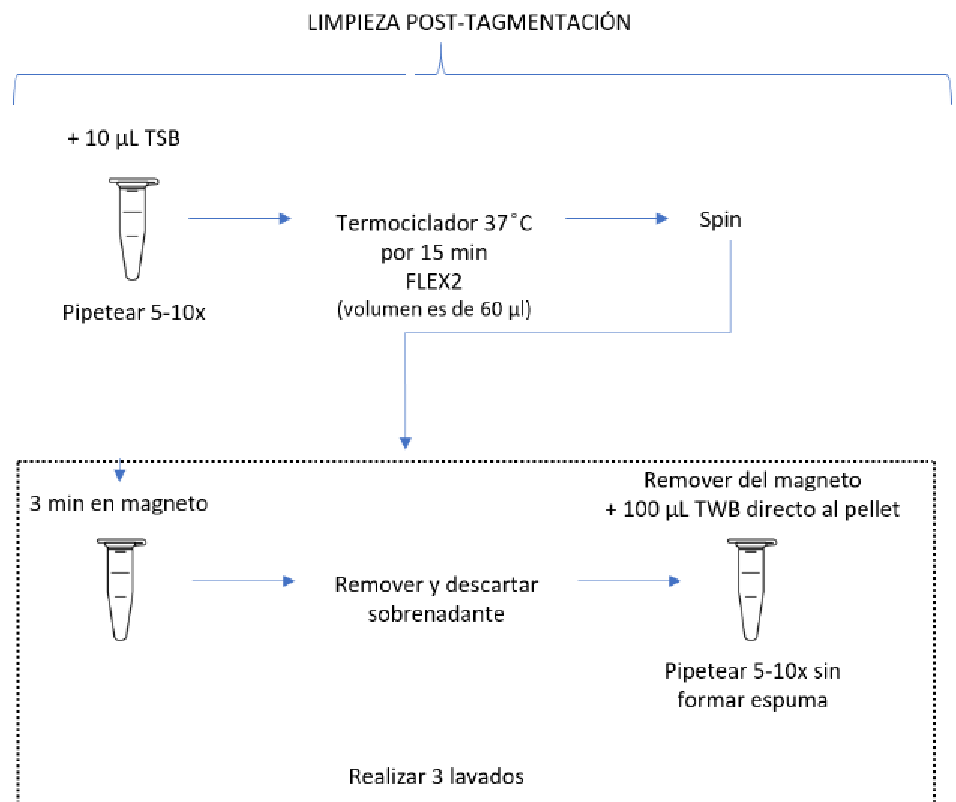

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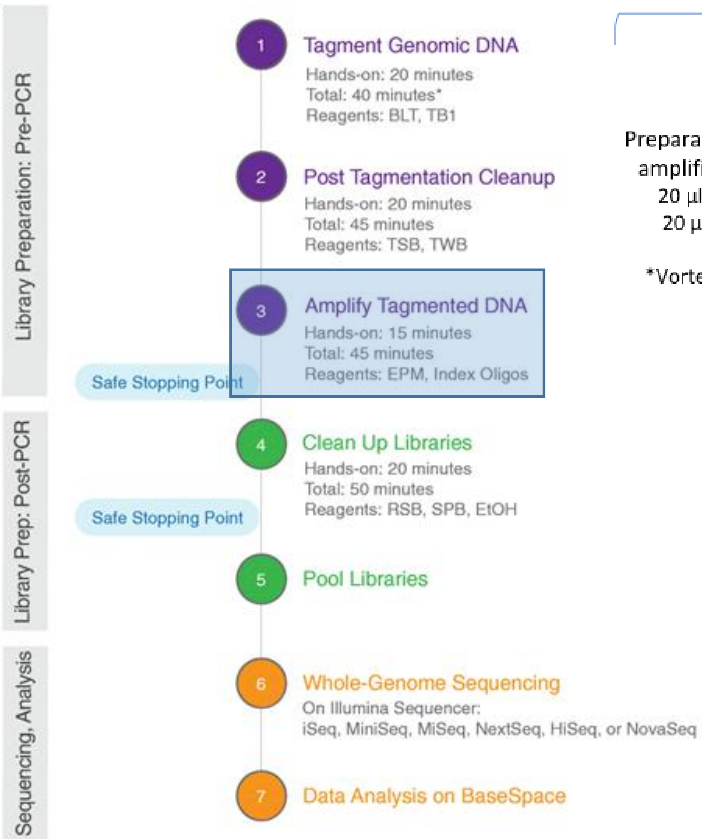


AMPLIFICACIÓN

Preparar Mix de amplificación
20 µl EPM
20 µl H_20

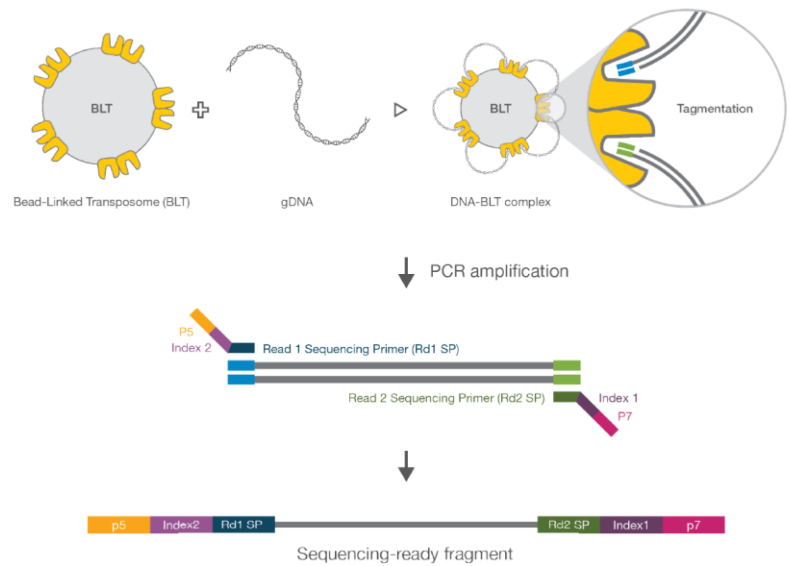
*Vortex EPM

→ Remover TWB y eliminar exceso

→ Remover del magneto + 40 µL MasterMix
Pipetear 5-10x

→ + 10 µL índices
Pipetear 5-10x

→ Termociclador ELEX3*
* Punto seguro para detener protocolo 2-8˚C
(volumen es de 50 µl)



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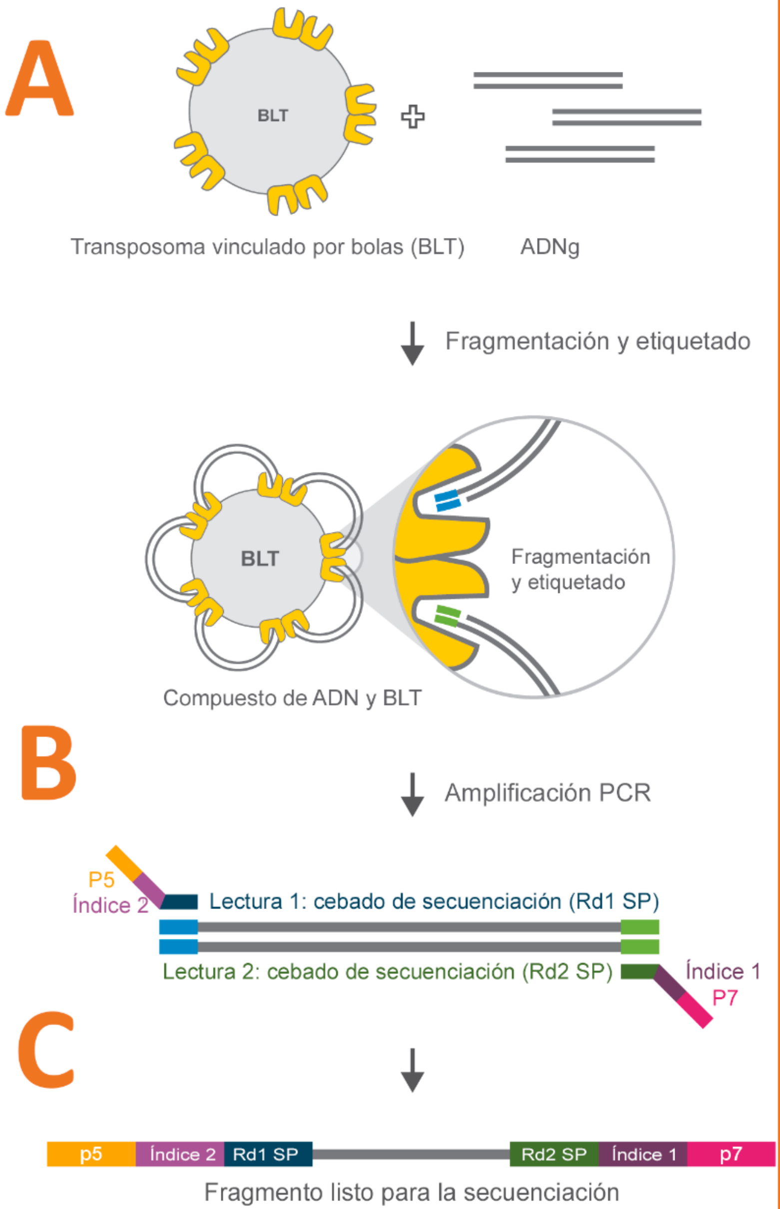


Procesos químicos de transposomas vinculados por bolas Nextera.

A

Los transposomas vinculados por bolas actúan como mediadores entre la fragmentación simultánea de ADNg y la adición de cebadores de secuenciación

B

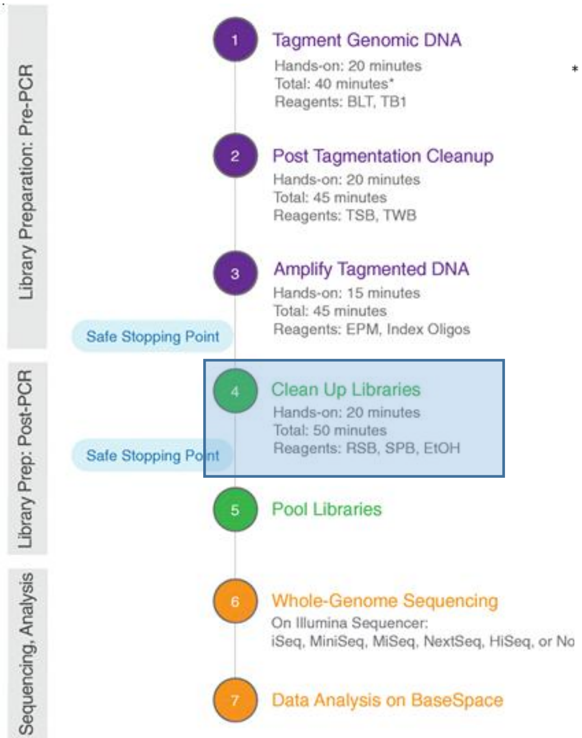
La PCR de ciclo reducido amplifica los fragmentos de ADN listos para la secuenciación y añade índices y adaptadores.

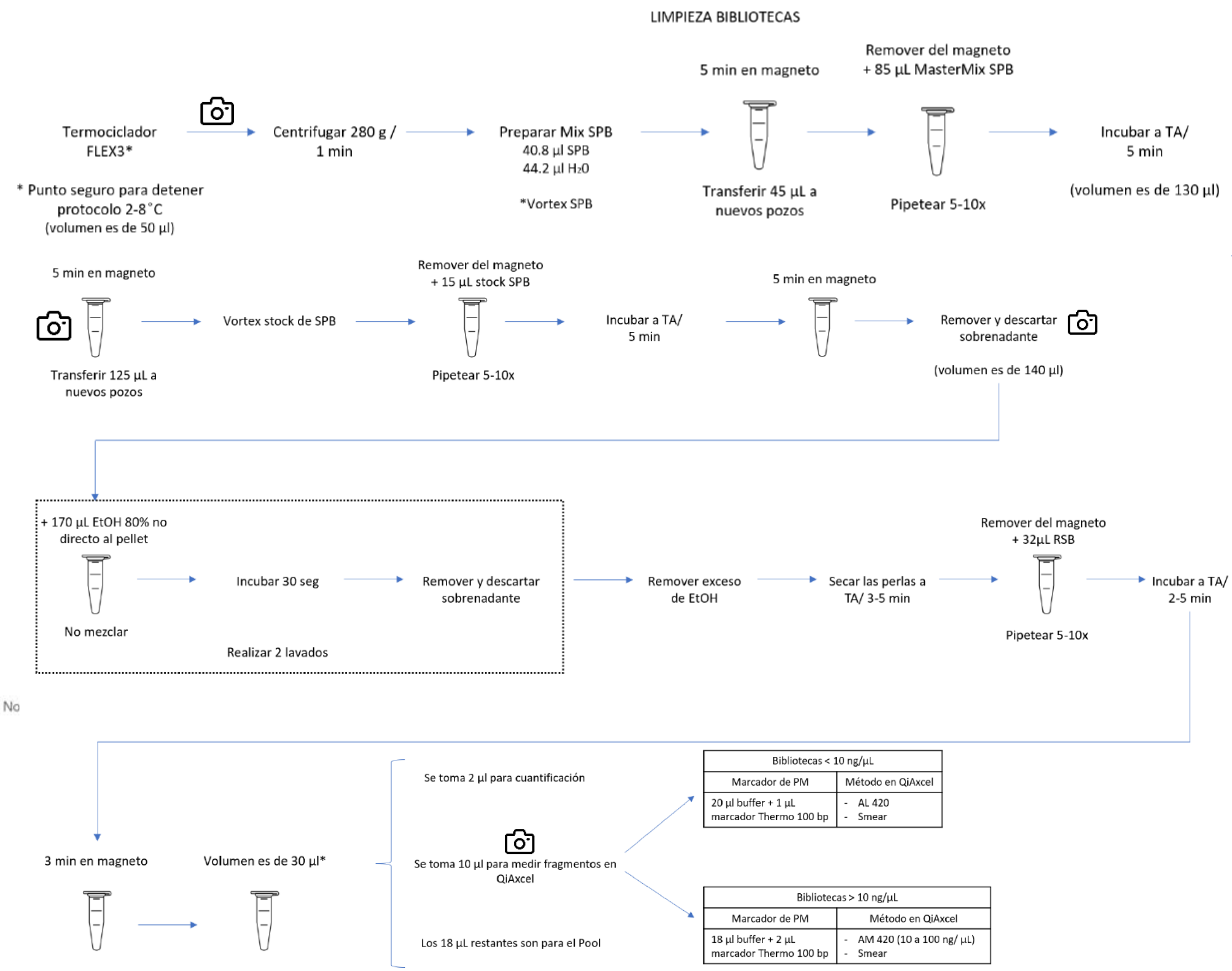
C

Los fragmentos listos para la secuenciación se lavan y se agrupan.

https://www.illumina.com/techniques/sequencing/ngs-library-prep/tagmentation.html

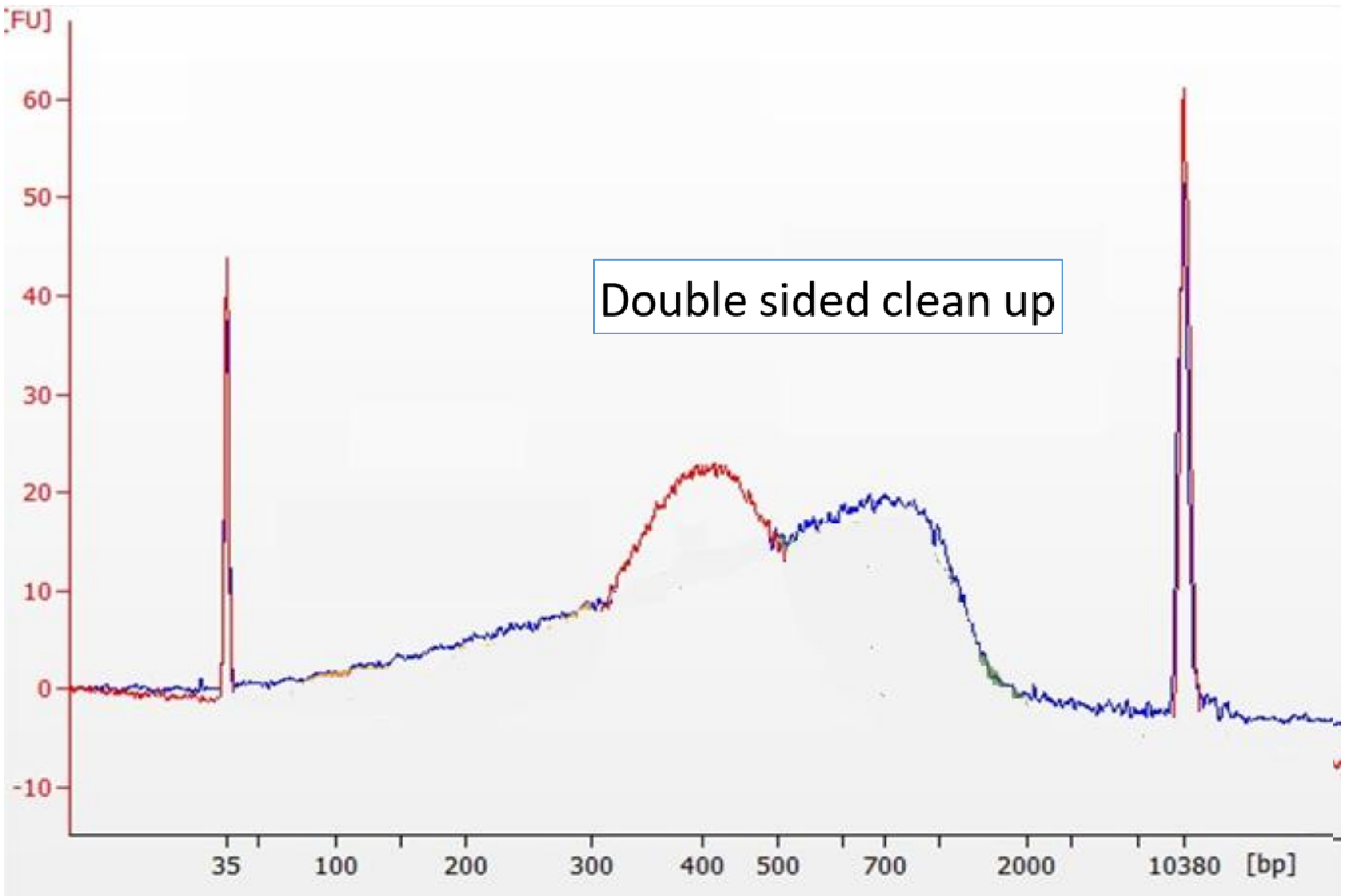
Preparación de bibliotecas para secuenciación: Illumina DNAprep kit



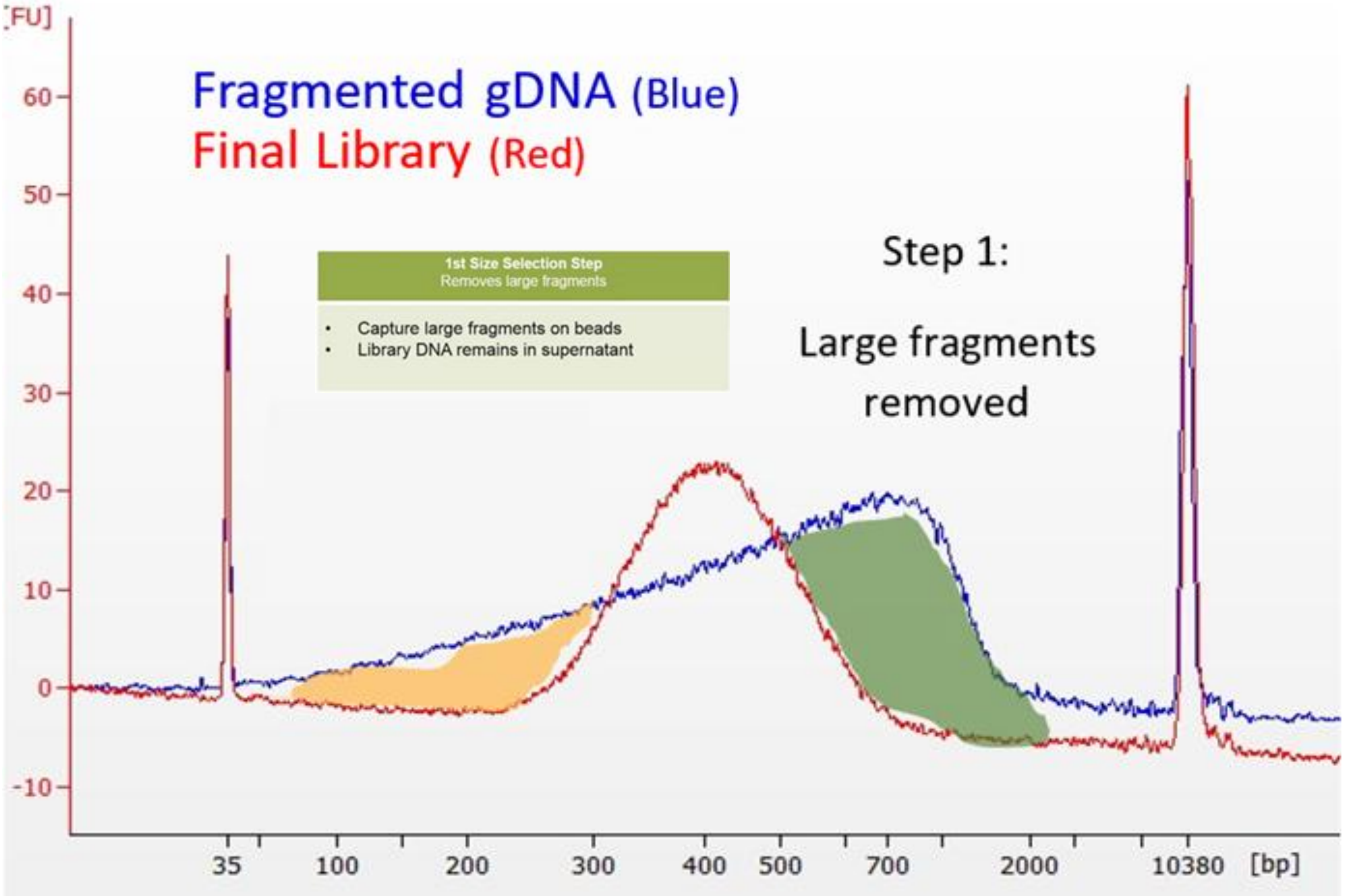


Bibliotecas < 10 ng/µL	
Marcador de PM	Método en QiAxcel
20 µl buffer + 1 µL marcador Thermo 100 bp	- AL 420 - Smear

Bibliotecas > 10 ng/µL	
Marcador de PM	Método en QiAxcel
18 µl buffer + 2 µL marcador Thermo 100 bp	- AM 420 (10 a 100 ng/ µL) - Smear



https://support.illumina.com/bulletins/2020/07/library-size-selection-using-sample-purification-beads.html



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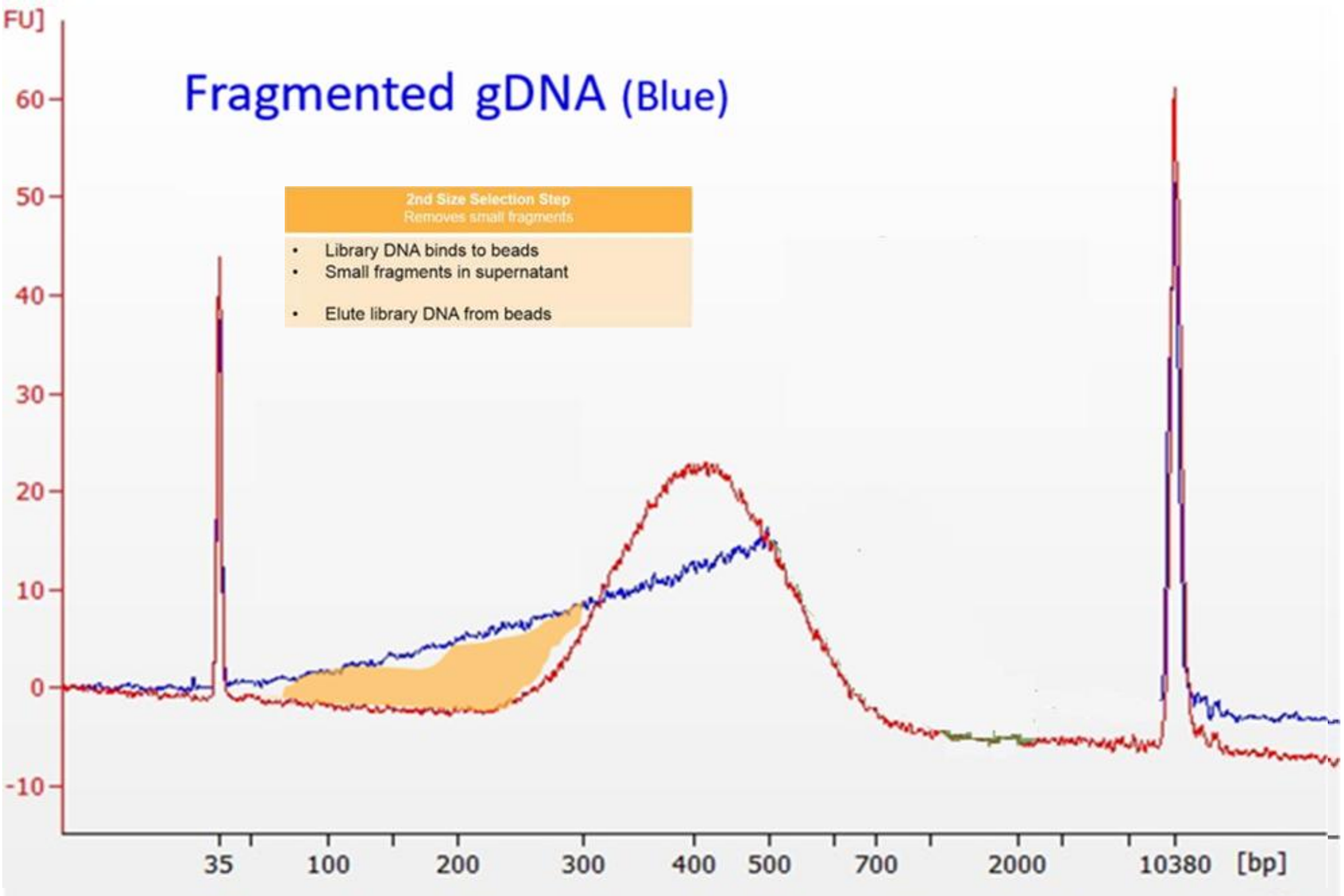

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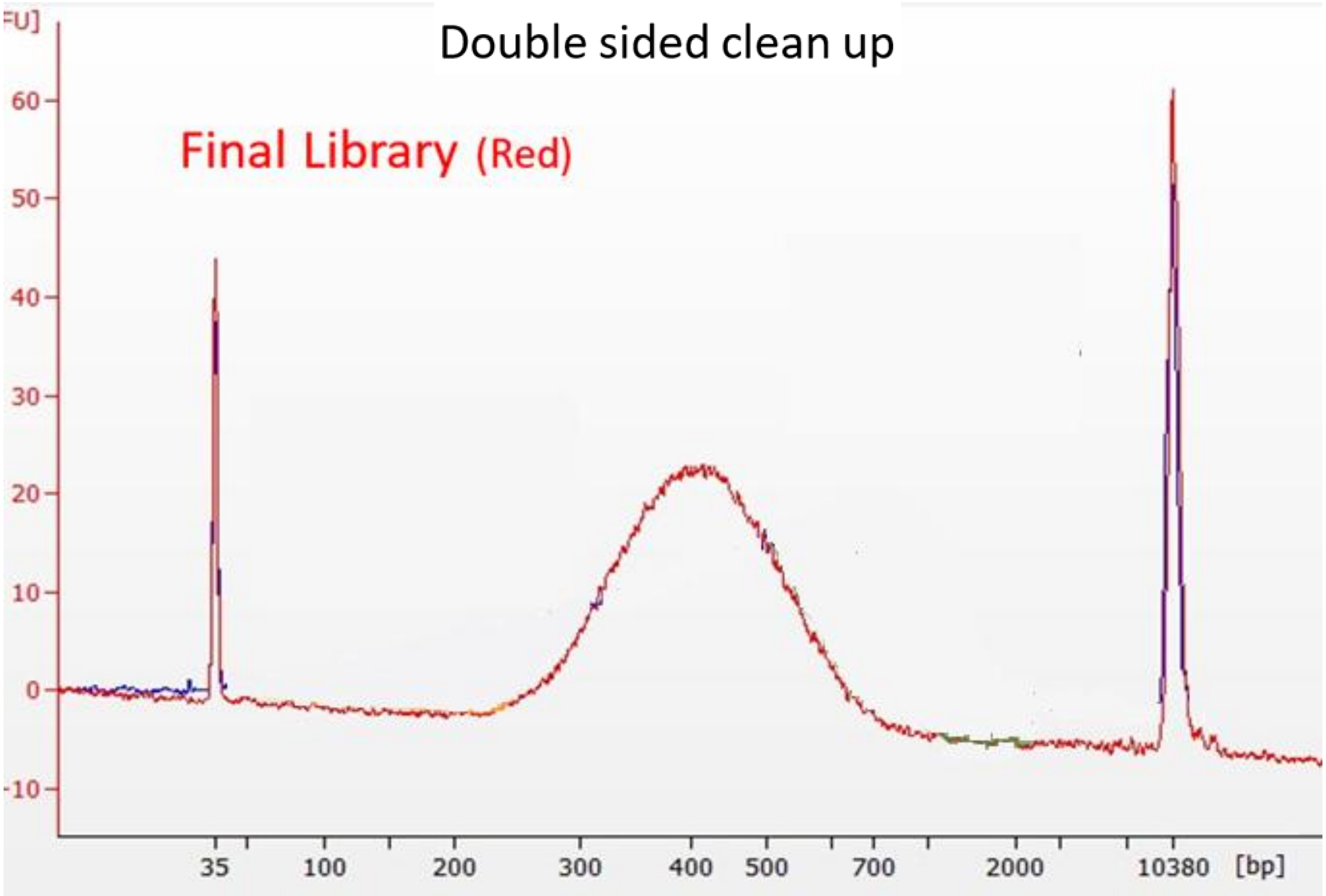

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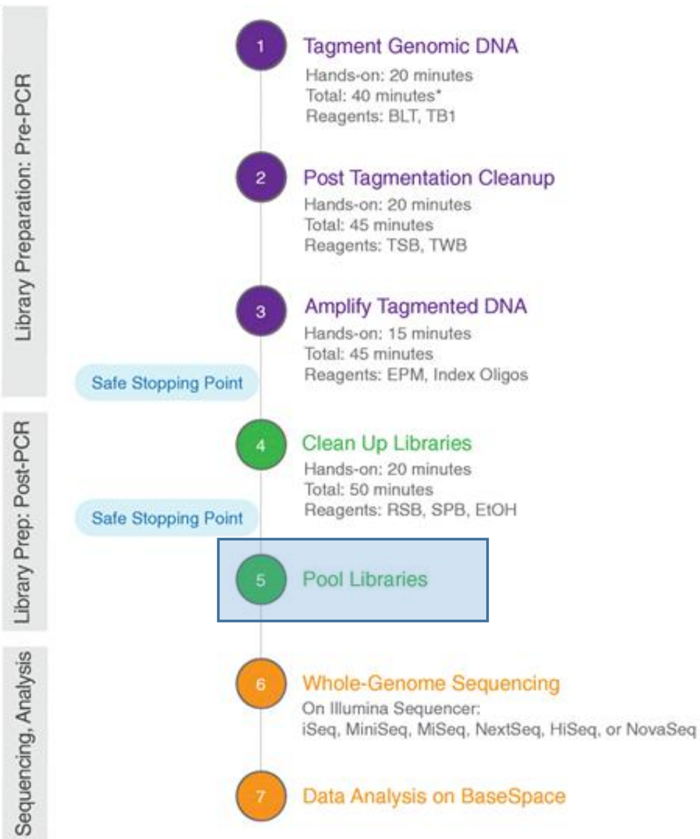

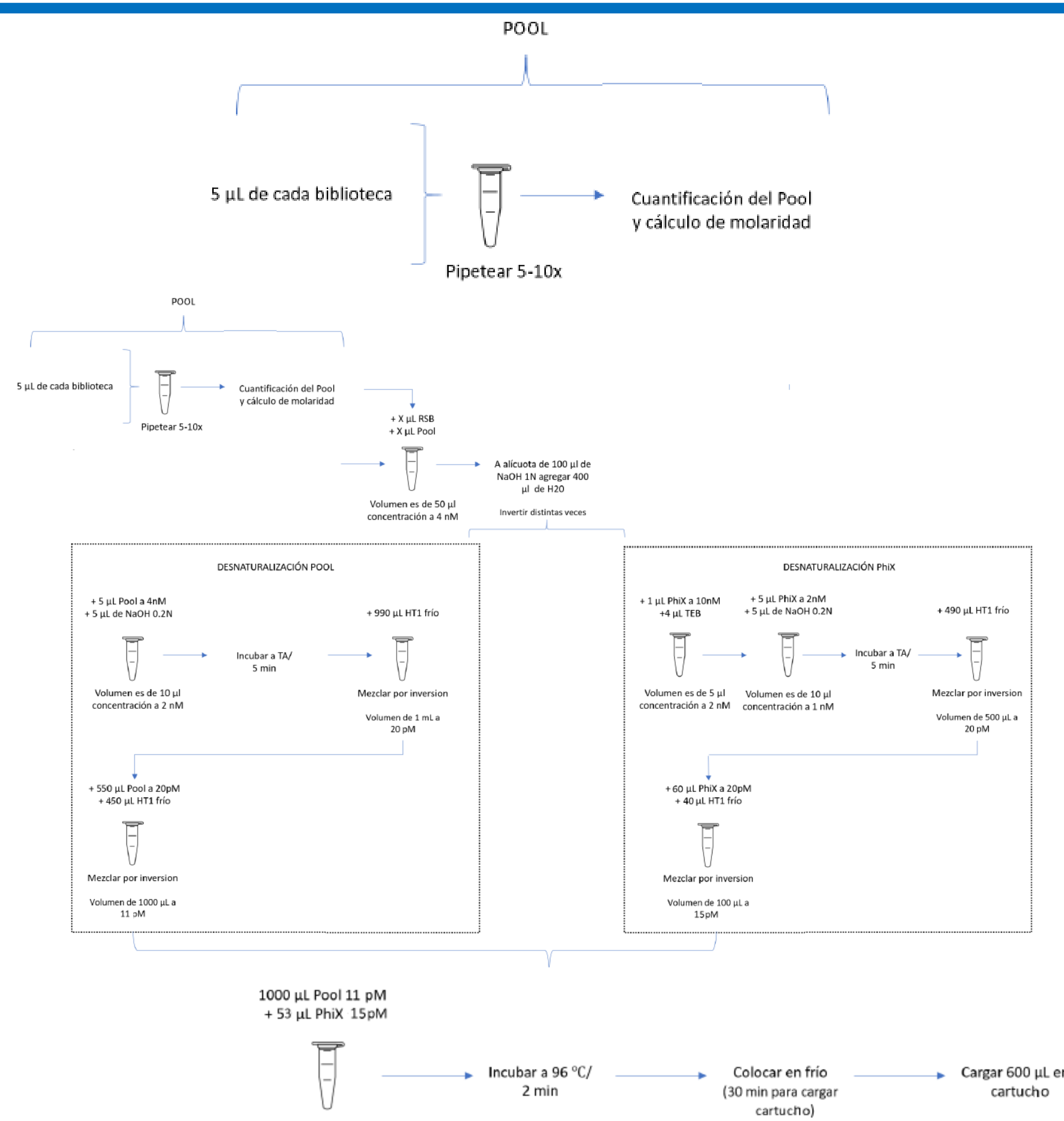

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Secuenciación por síntesis: Illumina

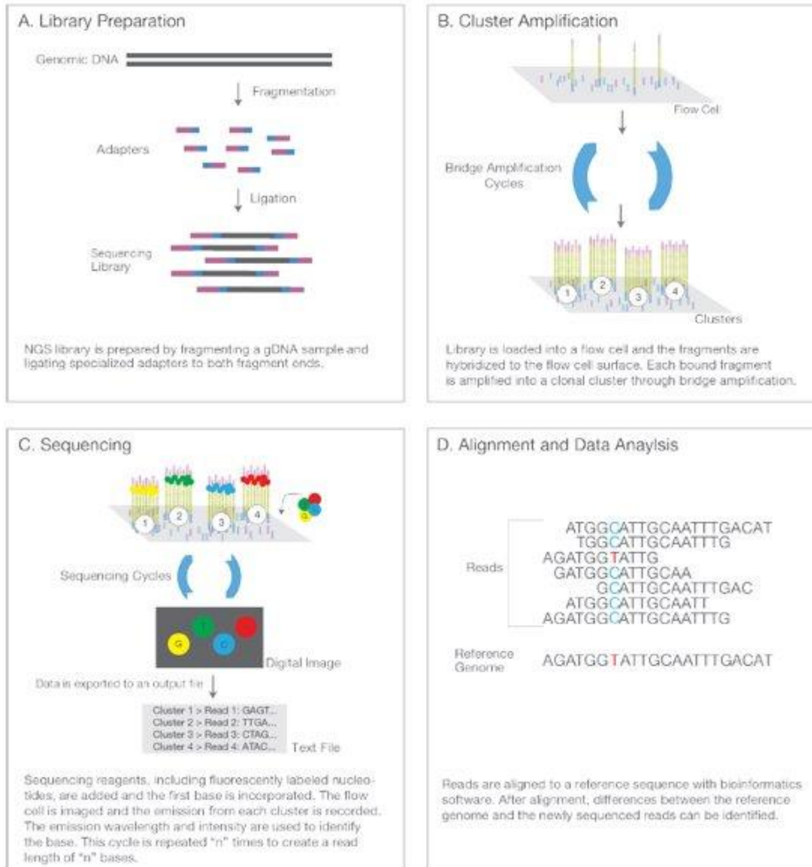
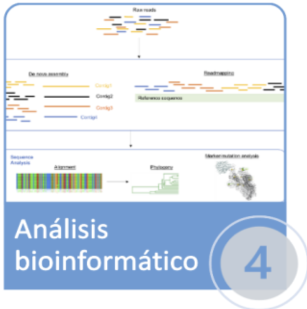


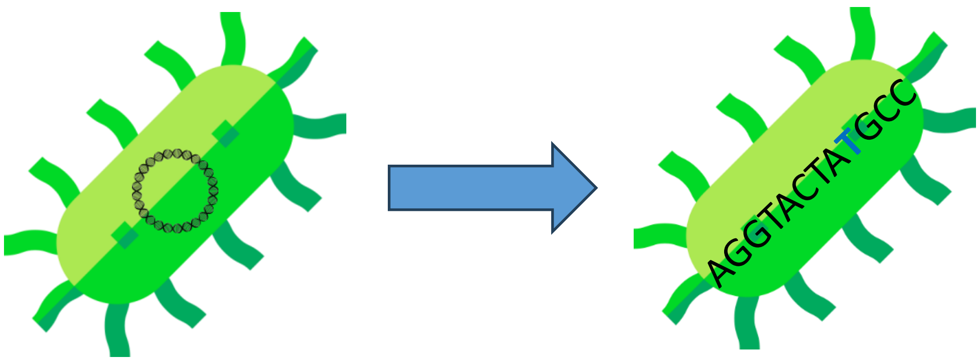
Figure 3: Next-Generation Sequencing Chemistry Overview—Illumina NGS includes four steps: (A) library preparation, (B) cluster generation, (C) sequencing, and (D) alignment and data analysis.

Bruinsma, S., Burgess, J., Schlingman, D. *et al.* Bead-linked transposomes enable a normalization-free workflow for NGS library preparation. *BMC Genomics* **19,** 722 (2018). https://doi.org/10.1186/s12864-018-5096-9

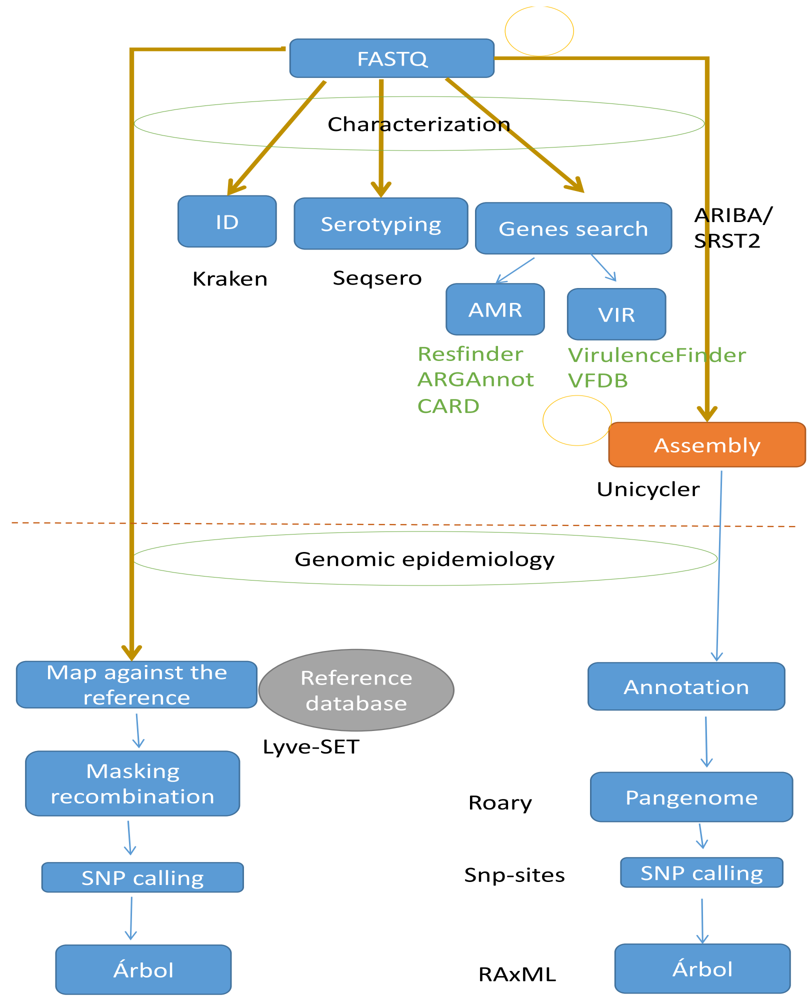
https://www.youtube.com/watch?v=fCd6B5HRaZ8&t=47s



¿Qué obtenemos y para qué nos sirve?

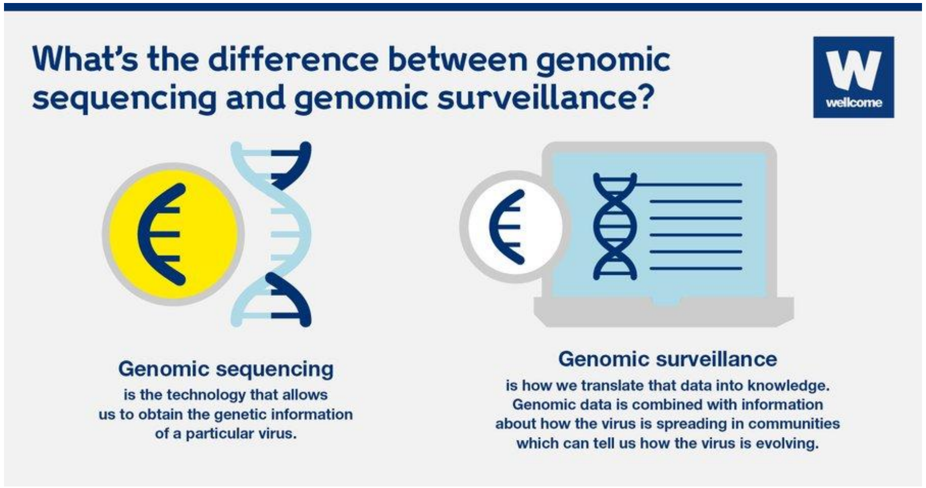


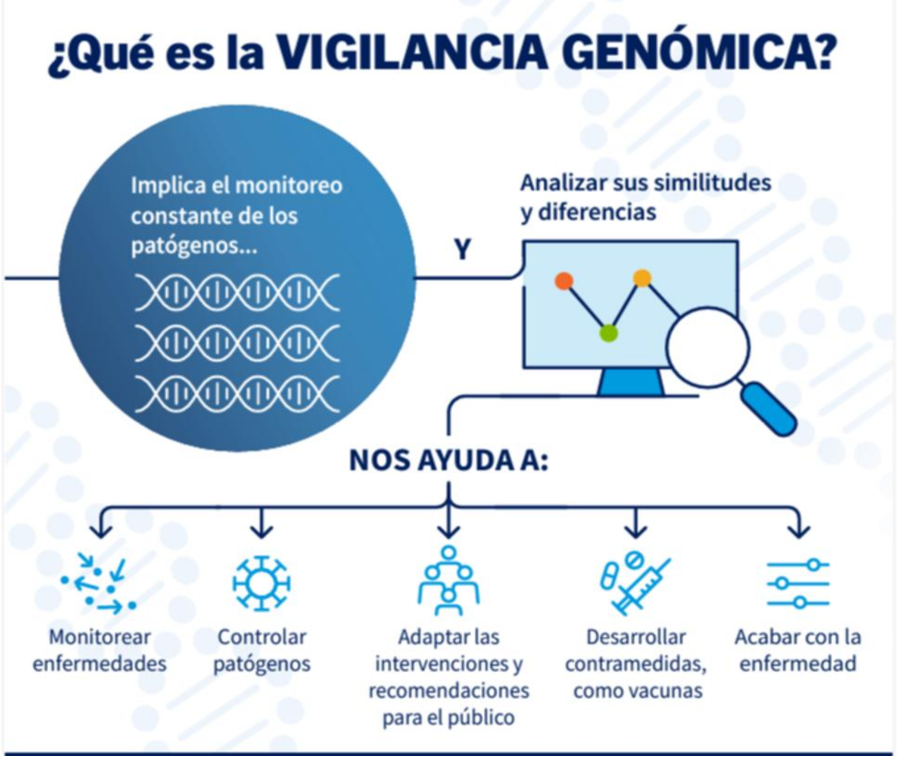
- Confirmar ID
- Tipificación y subtipificación
- Resistencia y virulencia
- Comparaciones genéticas



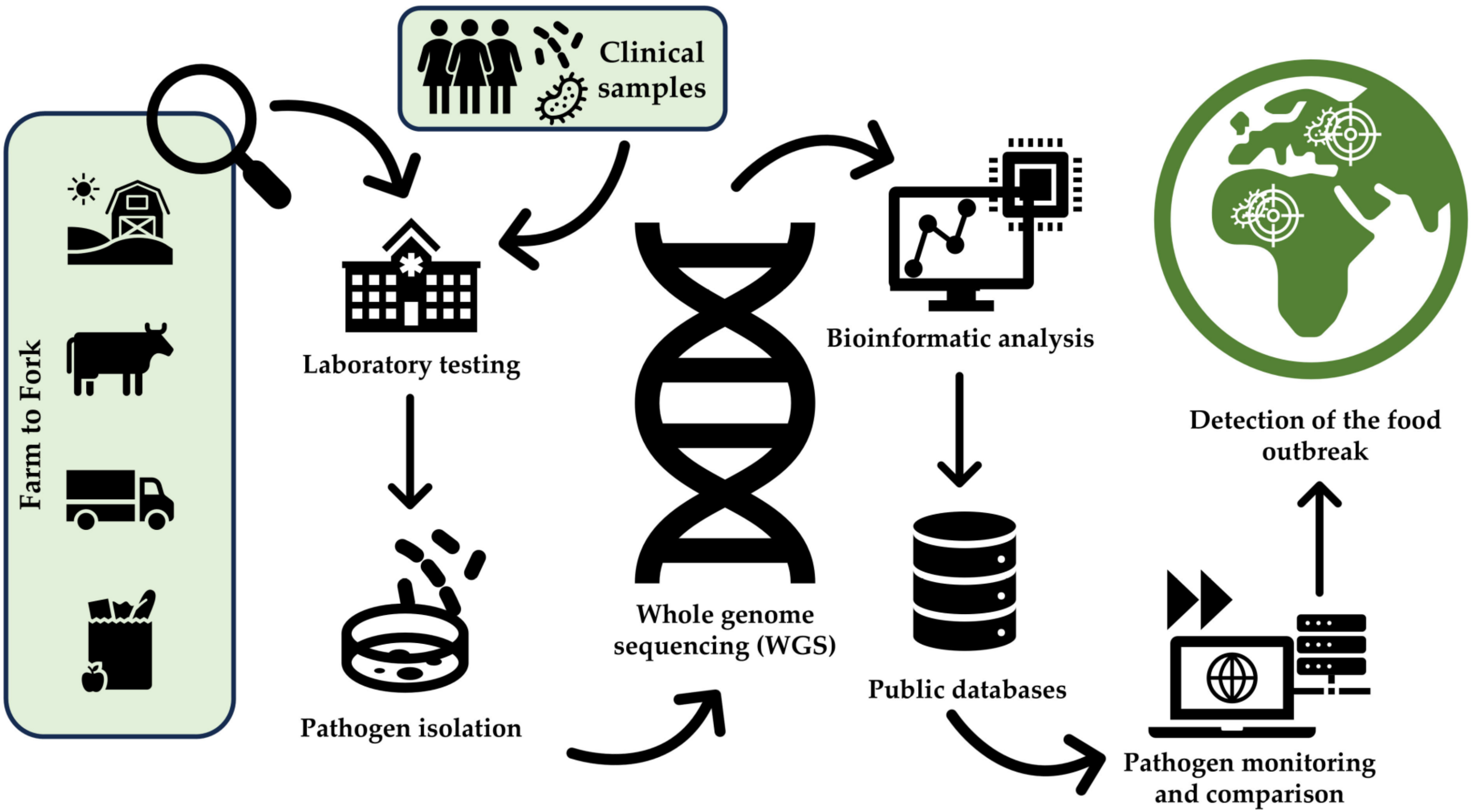
Vigilancia genómica

- Secuenciación genómica vrs Vigilancia genómica





Fundamentos de la vigilancia genómica aplicada a ETA



Gomes E, Araújo D, Nogueira T, Oliveira R, Silva S, Oliveira LVN, Azevedo NF, Almeida C and Castro J (2025) Advances in whole genome sequencing for foodborne pathogens: implications for clinical infectious disease surveillance and public health. Front. Cell. Infect. Microbiol. 15:1593219. doi: 10.3389/fcimb.2025.1593219

Agradecimientos

"La genómica no sustituye al trabajo de campo, lo potencia: nos da un mapa genético que, combinado con la epidemiología, permite actuar con más rapidez y precisión para proteger la salud pública."

INCIENSA
Instituto Costarricense de Investigación y Enseñanza en Nutrición y Salud

Laboratorio Nacional de Vigilancia
Secretaria de Salud